

CSCA January

1. Let $A = \{6, 7, 8, 9, 10\}$ be a set, and Φ to denote the empty set. Then which of the following statements is correct?
 - (a) $9 \in A$
 - (b) $\{6\} \in A$
 - (c) $6 \subseteq A$
 - (d) $\Phi \in A$
2. Let $A = \{x \mid -2 \leq x \leq 2\}$ and $B = \{x \mid x > 0\}$ be two sets. Then $A \cup B =$
 - (a) $\{x \mid x \leq -2\}$
 - (b) $\{x \mid 0 < x \leq 2\}$
 - (c) $\{x \mid x \geq -2\}$
 - (d) $\{x \mid -2 \leq x < 0\}$
3. The solution set of the inequality $x^2 - x - 2 > 0$ is ().
 - (a) $\{x \mid x < -1 \text{ or } x > 2\}$
 - (b) $\{x \mid -1 < x < 2\}$
 - (c) $\{x \mid x < -2 \text{ or } x > -1\}$
 - (d) $\{x \mid -2 < x < 1\}$
4. Let $\{a_n\}$ be an arithmetic sequence, with the first term $a_1 = 2$ and common difference $d = 3$. Then $a_{100} = ()$.
 - (a) 299
 - (b) 302
 - (c) 305
 - (d) 298
5. The domain of the function $f(x) = \frac{1}{x} + \sqrt{1-x}$ is ().
 - (a) $(-\infty, 1]$
 - (b) $(-\infty, 0) \cup (0, 1)$
 - (c) $(-\infty, 0) \cup (0, 1]$
 - (d) $[1, +\infty)$

6. Let $P(1, 2)$ be a point in the rectangular coordinate system. If a point Q together with P are symmetrical about the x axis, then the coordinates of Q are ().
- (a) $(2, 1)$
 - (b) $(1, -2)$
 - (c) $(-1, 2)$
 - (d) $(-1, -2)$
- 7.
8. About the function $f(x) = x^4 + 3$, which of the following statements is correct? ().
- (a) The function is odd
 - (b) None of the above
 - (c) The function is neither odd nor even
 - (d) The function is even
9. The inverse function of the function $y = x^3 + 3$, $x \in \mathbb{R}$ is ().
- (a) $y = \sqrt[3]{x-3}$, $x \in \mathbb{R}$
 - (b) $y = \sqrt[3]{x+3}$, $x \geq -3$
 - (c) $y = \sqrt[3]{x+3}$, $x \in \mathbb{R}$
 - (d) $y = \sqrt[3]{x-3}$, $x \geq 3$
10. Which of the following points is in the third quadrant? ().
- (a) $(-1, 2)$
 - (b) $(-1, -2)$
 - (c) $(1, 2)$
 - (d) $(1, -2)$
11. Given that $y = |x|$, which of the following conclusions is correct? ().
- (a) When $x < 0$, the function is an increasing function
 - (b) When $x > 0$, the function is an increasing function
 - (c) When $x \in \mathbb{R}$, the function is an decreasing function
 - (d) When $x \in \mathbb{R}$, the function is an increasing function

12. The solution set of the rational inequality $\frac{2x+1}{x-2} \leq 0$ is ().
- (a) $\left[-\frac{1}{2}, 2\right)$
 - (b) $\left(-\infty, -\frac{1}{2}\right] \cup (2, +\infty)$
 - (c) $\left[-\frac{1}{2}, 2\right]$
 - (d) $\left(-\infty, -\frac{1}{2}\right]$
13. Given that $a = 2 - \sqrt{3}$ and $b = 2 + \sqrt{3}$, the arithmetic mean of a and b is ().
- (a) 1
 - (b) 2
 - (c) ± 1
 - (d) -1
14. If the straight line l passes through the points $A(-2, 3)$ and $B(3, 1)$, then the slope of l is ().
- (a) $\frac{2}{5}$
 - (b) $\frac{5}{2}$
 - (c) $-\frac{2}{5}$
 - (d) $-\frac{5}{2}$
15. If α is the angle between the x -axis and a line containing the point $P(1, 3)$, then $\sin \alpha =$ ().
- (a) 3
 - (b) $\frac{3\sqrt{10}}{10}$
 - (c) $\frac{1}{3}$
 - (d) $\frac{\sqrt{10}}{10}$
16. Among the four sequences below, the number of arithmetic sequences is ().
- (1) 7, 13, 19, 25
 - (2) 2, 4, 7, 11
 - (3) $-1, -3, -5, -7$
 - (4) 2, 4, 8, 16

- (a) 2
 - (b) 3
 - (c) 1
 - (d) 4
17. The distance from a point $A(3, -2)$ to $B(-5, -1)$ on the plane is $|AB| = (\quad)$.
- (a) 65
 - (b) $\sqrt{65}$
 - (c) 13
 - (d) $\sqrt{13}$
18. If a straight line l has an angle of inclination of 45° , and it passes through the point $(0, 2)$, then the equation of l is $(\quad)$.
- (a) $y = 2x - 2$
 - (b) $y = x + 2$
 - (c) $y = 2x + 2$
 - (d) $y = x - 2$
19. If the equation of a circle is $x^2 + y^2 - 4x - 3 = 0$, then the center and radius are given by $(\quad)$.
- (a) $(0, 2)$ and $\sqrt{7}$
 - (b) $(0, 2)$ and 7
 - (c) $(2, 0)$ and $\sqrt{7}$
 - (d) $(2, 0)$ and 7
20. Given that $\sin \alpha = \frac{4}{5}$, and the angle α belongs to the second quadrant, then which of the following statements is correct? $(\quad)$
- (a) $\cos \alpha = \frac{3}{5}$
 - (b) $\cos \alpha = -\frac{3}{5}$
 - (c) $\tan \alpha = -\frac{3}{4}$
 - (d) $\tan \alpha = \frac{4}{3}$

21. In an arithmetic sequence $\{a_n\}$, if $a_2 = 1$ and $a_4 = 5$, then the first term a_1 and common difference d are () respectively.
- (a) $-1, 2$
 - (b) $1, -2$
 - (c) $-1, -2$
 - (d) $1, 2$
22. Let a, b, c be three real numbers. Which of the following statements is correct? ()
- (a) If $a > b$, then $ac > bc$
 - (b) If $a > b$, then $a + c > b + c$
 - (c) If $ac > bc$, then $a > b$
 - (d) If $a > b$, then $a^2 > b^2$
23. About the exponential function $y = a^x$ ($a > 0$, and $a \neq 1$), which of the following statements is incorrect? ()
- (a) The range is $(0, +\infty)$
 - (b) The graph passes through the point $(0, 1)$
 - (c) The function is increasing over the domain
 - (d) The domain is $\mathbb{R}$
24. If $\sin \alpha = \frac{3}{5}$, and α belongs to the first quadrant, then $\sin 2\alpha = ()$.
- (a) $\frac{12}{25}$
 - (b) $\frac{24}{25}$
 - (c) $\frac{18}{25}$
 - (d) $\frac{7}{25}$
25. The intersection of the lines $l_1 : 3x - y + 8 = 0$ and $l_2 : x + 2y - 9 = 0$ has coordinates ().
- (a) $(-5, 1)$
 - (b) $(1, -5)$
 - (c) $(5, -1)$
 - (d) $(-1, 5)$

26. Given that the equation of a hyperbola is $\frac{x^2}{64} - \frac{y^2}{16} = 1$, which of the following statements is correct? ()
- (a) The focal distance is $8\sqrt{5}$
 - (b) The semi-major axis has length 4
 - (c) The eccentricity of the hyperbola is $\sqrt{5}$
 - (d) The semi-minor axis has length 8
27. Which of the following inequalities is correct? ()
- (a) $2.1^{-2} > 1.2^{-2}$
 - (b) $0.75^{-0.2} > 0.75^{-0.4}$
 - (c) $3^{2.25} > 3^3$
 - (d) $2.1^{\frac{2}{3}} > 1.2^{\frac{2}{3}}$
28. A circle has center at $(-3, 2)$ and radius 4. Then the equation of the circle is ().
- (a) $(x + 3)^2 + (y - 2)^2 = 4$
 - (b) $(x - 3)^2 + (y + 2)^2 = 4$
 - (c) $(x - 3)^2 + (y + 2)^2 = 16$
 - (d) $(x + 3)^2 + (y - 2)^2 = 16$
29. Let $a > 0$, and $b > 0$ be real numbers. Then which of the following statements is incorrect? ()
- (a) $\ln(a^b) = b \ln a$
 - (b) $\log_a b = \frac{\ln a}{\ln b} \quad (a \neq 1, b \neq 1)$
 - (c) $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$
 - (d) $\ln(ab) = \ln a + \ln b$
30. Given the sine function $y = \sin x$, which of the following statements is incorrect? ()
- (a) The maximum and minimum values are 1 and -1 respectively
 - (b) The function is periodic function
 - (c) The function is even
 - (d) The domain is $\mathbb{R}$

31. Let the focus of the parabola $y^2 = 4x$ be F . Suppose that a point P is on the parabola, and the horizontal coordinate of P is 4. Then $|PF| = ()$.
- (a) 2
 - (b) 3
 - (c) 4
 - (d) 5
32. Given two straight lines $l_1 : ax + (a - 1)y + 3 = 0$ and $l_2 : 2x + ay - 1 = 0$. If $l_1 \perp l_2$, then the value of a is $()$.
- (a) -1
 - (b) -1 or 1
 - (c) 1
 - (d) 0 or -1
33. The distance between the points $P(-1, 2)$ and $Q(3, 1)$ is $()$.
- (a) $\sqrt{5}$
 - (b) $\sqrt{17}$
 - (c) $\sqrt{13}$
 - (d) 5
34. Let $\cos \alpha = -\frac{\sqrt{5}}{5}$, and $\alpha \in \left(\frac{\pi}{2}, \pi\right)$. Then $\sin\left(\frac{\pi}{6} + \alpha\right) = ()$.
- (a) $\frac{2\sqrt{5} - \sqrt{15}}{10}$
 - (b) $\frac{2\sqrt{15} + \sqrt{5}}{10}$
 - (c) $\frac{2\sqrt{15} - \sqrt{5}}{10}$
 - (d) $\frac{2\sqrt{5} + \sqrt{15}}{10}$
35. Which of the following statements is correct? $()$
- (a) $y = (\sqrt{x})^2$ and $y = \sqrt{x^2}$ are the same function
 - (b) $y = \frac{x^4 - 1}{x^2 + 1}$ and $y = x^2 - 1$ are the same function
 - (c) $y = 1$ and $y = x^2$ are the same function
 - (d) $y = x$ and $y = \sqrt{x^2}$ are the same function

36. Let $\sin \alpha = -\frac{12}{13}$, and $\alpha \in \left(\pi, \frac{3}{2}\pi\right)$. Then $\cos \frac{\alpha}{2} = ()$.

- (a) $\frac{3\sqrt{13}}{13}$
- (b) $-\frac{3\sqrt{13}}{13}$
- (c) $-\frac{2\sqrt{13}}{13}$
- (d) $\frac{2\sqrt{13}}{13}$

37. Suppose that $\cos \alpha = -\frac{1}{2}$, $\alpha \in \left(\frac{\pi}{2}, \pi\right)$, then the value of $\sin \frac{\alpha}{2}$ is $()$.

- (a) $-\frac{\sqrt{3}}{2}$
- (b) $\frac{\sqrt{3}}{2}$
- (c) $-\frac{1}{2}$
- (d) $\frac{1}{2}$

38. Which of the following statements is correct? $()$

- (a) $\sin(\pi + \alpha) = \sin \alpha$
- (b) $\cos(\pi + \alpha) = \cos \alpha$
- (c) $\sin(-\alpha) = \sin \alpha$
- (d) $\cos(-\alpha) = \cos \alpha$

39. About the tangent function $y = \tan x$, which of the following statements is correct? $()$

- (a) The range is $[-1, 1]$
- (b) The domain is $\mathbb{R}$
- (c) The function is even
- (d) The minimum positive period of the function is π

40. The general formula for the sequence $-\frac{1}{5}, \frac{1}{7}, -\frac{1}{9}, \frac{1}{11}, \dots$ is $a_n = ()$.
- (a) $\frac{(-1)^{n-1}}{3n+2}$
 - (b) $\frac{(-1)^n}{2n+3}$
 - (c) $\frac{(-1)^{n-1}}{2n+3}$
 - (d) $\frac{(-1)^n}{3n+2}$
41. If the graph of the function $f(x) = 2 + \log_a(x-3)$ ($a > 0$, and $a \neq 1$) passes through a fixed point P , then the coordinates of P are $()$.
- (a) $(3, 1)$
 - (b) $(3, 0)$
 - (c) $(4, 0)$
 - (d) $(4, 2)$
42. The equation of the directrix of the parabola $y^2 = -x$ is $()$.
- (a) $x = \frac{1}{4}$
 - (b) $x = -\frac{1}{4}$
 - (c) $x = -\frac{1}{2}$
 - (d) $x = \frac{1}{2}$
43. Suppose that the foci of the ellipse $\frac{x^2}{4} + \frac{y^2}{2} = 1$ are F_1, F_2 , and the point P is on the ellipse. Then $|PF_1| + |PF_2| = ()$.
- (a) 2
 - (b) 4
 - (c) 8
 - (d) 6

44. Given that $\overrightarrow{AB} = \vec{a} + 5\vec{b}$, $\overrightarrow{BC} = -2\vec{a} + 8\vec{b}$, $\overrightarrow{CD} = 3\vec{a} - 3\vec{b}$. Thus, ().
- (a) A, C, D are on the same line
 - (b) A, B, D are on the same line
 - (c) B, C, D are on the same line
 - (d) A, B, C are on the same line
45. Suppose that a straight line l is perpendicular to another line $l_1 : x + 2y + 4 = 0$, and passes through the intersection of $l_2 : x + y + 1 = 0$ and $l_3 : 2x + y - 1 = 0$. The equation of l is ().
- (a) $2x - y - 7 = 0$
 - (b) $2x + y - 5 = 0$
 - (c) $x - 2y - 1 = 0$
 - (d) $x + 2y - 3 = 0$
46. On the complex plane, the point representing a complex number z is on the straight line $x - y = 0$. If z is a root of a quadratic equation $x^2 + mx + 4 = 0$, then $m = ()$.
- (a) $\sqrt{2}$ or $2\sqrt{2}$
 - (b) $2\sqrt{2}$ or $-2\sqrt{2}$
 - (c) $-\sqrt{2}$ or $-2\sqrt{2}$
 - (d) $\sqrt{2}$ or $-\sqrt{2}$
47. Suppose that, in a sequence $\{a_n\}$, we have $a_1 = 1$, and $\frac{1}{a_n} + \frac{2}{a_{n+1}} = 0$ ($n = 1, 2, \dots$). Suppose that another sequence $\{b_n\}$ satisfies $b_n = |a_n|$ ($n = 1, 2, \dots$). Then, the sum of the first n terms of the sequence $\{b_n\}$ is $S_n = ()$.
- (a) $\frac{1 + 2^n}{3}$
 - (b) $\frac{|1 - (-2)^n|}{3}$
 - (c) $2^n - 1$
 - (d) $(-2)^n - 1$

48. In daily life, people enjoy using running apps to track and share their exercise routes. To understand the usage of running apps among high school and college students in a certain region, 200 high school students and 80 college students were randomly chosen, and their favorite running apps were recorded as follows:

	App A	App B	App C	App D
High school students	80	40	60	20
College students	30	20	20	10

Assuming that college students' and high school students' preferences for running apps are independent of each other, if one person is randomly selected from each group, the probability that their favorite running apps are different is ().

- (a) $\frac{23}{80}$
- (b) $\frac{57}{80}$
- (c) $\frac{9}{20}$
- (d) $\frac{11}{20}$